

Shipping Route Optimization Using Data Analytics: An Interactive Streamlit Dashboard for Logistics Performance Analysis

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ABSTRACT

The rapid growth of e-commerce and nationwide product distribution has significantly increased the importance of efficient logistics and supply chain management. Shipping delays, inefficient transportation routes, and poor visibility into operational performance can adversely affect customer satisfaction and increase overall logistics costs. This project presents a data-driven approach to analyse and optimise shipping operations for Nassau Candy Distributor by developing an interactive analytics dashboard using Python and Streamlit.

The project utilises historical shipment data to evaluate logistics performance through key performance indicators such as Shipping Lead Time, Gross Profit, Sales, Route Efficiency Score, and Delay Frequency. Data preprocessing techniques including cleaning, validation, and feature engineering were performed to ensure accurate analysis. Multiple interactive visualisations were developed using Plotly to identify shipment trends, regional performance, factory productivity, and shipping mode efficiency.

An interactive Streamlit dashboard was designed to enable users to filter data dynamically by region, division, ship mode, and product, allowing real-time business insights. The dashboard assists decision-makers in identifying operational bottlenecks, evaluating shipping performance, and supporting data-driven logistics optimisation.

The proposed solution demonstrates how business intelligence and data analytics techniques can improve supply chain visibility and operational decision-making. The developed dashboard provides a scalable framework that can be extended with predictive analytics and real-time data integration for future logistics optimisation.

Keywords:

Supply Chain Analytics, Logistics, Shipping Route Optimisation, Business Intelligence, Streamlit, Python, Data Visualisation, Plotly, Data Analytics.

I. INTRODUCTION

Modern supply chain management has evolved significantly with the increasing growth of e-commerce, global trade, and customer expectations for faster deliveries. Efficient logistics operations play a critical role in ensuring products are delivered on time while maintaining operational costs and customer satisfaction. Organisations handling nationwide product distribution generate vast amounts of logistics data, including shipment records, transportation routes, delivery times, sales, and inventory information. Analysing this data enables organisations to identify operational bottlenecks and optimise decision-making.

Data Analytics has become one of the most effective approaches for transforming raw logistics data into meaningful business insights. By applying data cleaning, feature engineering, statistical analysis, and interactive visualisation techniques, organisations can monitor shipping performance, evaluate transportation efficiency, and improve resource utilisation. Business Intelligence (BI) dashboards further enhance decision-making by presenting complex operational information in a simple, interactive, and visually appealing manner.

Nassau Candy Distributor is one of the leading confectionery distributors in the United States, delivering products from multiple manufacturing facilities to customers located across various states and regions. Due to the large geographical coverage and multiple shipping methods, maintaining efficient logistics operations is a challenging task. Variations in shipping lead time, transportation routes, and factory performance can directly influence operational efficiency and customer experience.

This project focuses on developing an interactive Shipping Route Optimisation Dashboard using Python, Pandas, Plotly, and Streamlit. Historical shipment data is analysed to measure logistics performance through Key Performance Indicators (KPIs) such as Shipping Lead Time, Gross Profit, Sales, Route Efficiency Score, and Delay Frequency. Interactive visualisations allow users to explore shipment performance across regions, divisions, factories, and shipping modes.

The developed dashboard enables logistics managers and business analysts to identify high-performing routes, detect operational bottlenecks, compare shipping methods, and monitor business performance through real-time filtering capabilities. The proposed solution demonstrates how modern data analytics techniques can support evidence-based decision-making and improve overall logistics efficiency.

Furthermore, the project highlights the practical application of Python-based analytics tools for solving real-world business problems. The combination of data preprocessing, exploratory data analysis, and dashboard development provides a scalable framework that can be extended to predictive analytics, machine learning models, and real-time shipment monitoring in future logistics systems.

II. PROBLEM STATEMENT

The rapid expansion of nationwide distribution networks has significantly increased the complexity of logistics operations.

Nassau Candy Distributor manages shipments from multiple manufacturing facilities to customers located across different regions of the United States. Although the organisation possesses extensive historical shipment data, logistics decisions are often made without a comprehensive understanding of route-level performance and operational efficiency.

One of the primary challenges faced by the organisation is the inability to identify consistently efficient shipping routes and detect routes experiencing frequent delivery delays. Variations in shipping lead time across regions, states, factories, and shipping methods make it difficult to evaluate logistics performance using traditional reporting techniques. Without detailed analytical insights, identifying operational bottlenecks and improving delivery efficiency becomes a reactive process rather than a proactive strategy.

Additionally, the organisation lacks an integrated platform that enables business users to monitor key logistics performance indicators such as shipping lead time, route efficiency, shipment volume, gross profit, and regional performance through interactive visualisations. The absence of such analytical capabilities limits data-driven decision-making and affects overall supply chain optimisation.

Therefore, there is a need to develop an interactive analytics solution that transforms raw shipment data into meaningful business intelligence. This project addresses this challenge by applying data analytics techniques, feature engineering, and interactive dashboard visualisation to evaluate shipping performance, identify operational bottlenecks, and support informed logistics decision-making for Nassau Candy Distributor.

III. OBJECTIVES

The primary objective of this project is to develop an interactive logistics analytics dashboard that enables data-driven evaluation of shipping performance for Nassau Candy Distributor.

The specific objectives of the project are:

- Analyse historical shipment and sales data to understand logistics performance.
- Calculate Shipping Lead Time and other logistics Key Performance Indicators (KPIs).
- Perform data cleaning, validation, and feature engineering to prepare the dataset for analysis.
- Identify the most efficient and least efficient shipping routes based on delivery performance.
- Compare the performance of different shipping modes across multiple regions.
- Analyse regional sales, gross profit, shipment volume, and factory productivity.
- Develop interactive visualisations using Plotly to simplify logistics analysis.
- Build a Streamlit dashboard with dynamic filtering capabilities for business users.
- Generate business insights and recommendations to support logistics optimisation and operational efficiency.

IV. DATASET DESCRIPTION

The dataset used in this project was provided as part of the Unified Mentor internship programme. It represents historical shipment and sales transactions for Nassau Candy Distributor, a nationwide confectionery distributor operating across multiple regions in the United States. The dataset contains order information, customer details, shipping records, sales performance, product information, and manufacturing factory locations.

The primary objective of the dataset is to analyse shipping efficiency, identify logistics bottlenecks, and evaluate business performance using historical shipment data. Each record represents an individual customer order that contains both business and logistics attributes.

The dataset consists of the following major categories of information:

- Order Information
 - Order ID
 - Order Date
 - Ship Date
 - Ship Mode
- Customer Information
 - Customer ID
 - City
 - State/Province
 - Region
 - Postal Code
 - Country/Region
- Product Information
 - Product ID
 - Product Name
 - Division
- Business Performance
 - Sales
 - Cost
 - Gross Profit
 - Units Sold

To support logistics analysis, additional information regarding manufacturing factories and product-to-factory relationships was provided. These factory locations were used to analyse shipment routes and evaluate transportation efficiency.

The project also incorporates factory geographical coordinates, enabling future enhancements such as route visualisation and geographical mapping.

A. Manufacturing Factory Locations

B. Product and Factory Correlation

Several additional analytical features were derived from the original dataset to improve logistics analysis and dashboard visualisation.

The engineered features include:

- Shipping Lead Time
Calculated as the difference between Ship Date and Order Date.

- **Route Efficiency Score**
A normalised performance score used to compare shipping routes based on average delivery time.
- **Delay Frequency**
Identification of delayed shipments based on lead-time thresholds.
- **Factory-to-Region Route**
Used to analyse logistics performance across different manufacturing facilities and customer regions.
- **Factory-to-State Route**
Created for detailed state-level shipping analysis.

These engineered variables significantly improved the quality of business insights generated through the Streamlit dashboard.

V. METHODOLOGY

The methodology adopted in this research follows a systematic data analytics workflow beginning with raw data preprocessing and ending with the development of an interactive analytics dashboard. The complete workflow is illustrated in Figure ??.

The project was divided into six major phases:

- 1) Data Collection
- 2) Data Cleaning and Validation
- 3) Feature Engineering
- 4) Exploratory Data Analysis
- 5) KPI Development
- 6) Dashboard Design and Visualisation

VI. DASHBOARD IMPLEMENTATION

The logistics analytics dashboard was developed using Streamlit, a Python-based framework for creating interactive data applications. The dashboard provides a user-friendly interface for analyzing shipping performance, sales trends, profitability, and route efficiency.

The application integrates data pre processing, exploratory data analysis, and visualization techniques to convert raw logistics data into meaningful business insights. Users can interact with different filters and charts to analyze specific regions, shipping modes, and performance indicators.

The dashboard includes key analytical components such as:

- Shipping lead time analysis
- Sales and profit performance visualization
- Regional performance comparison
- Route efficiency analysis
- Interactive filtering for business decision-making

Each phase is discussed in detail below.

A. Data Collection

The shipment dataset provided by Unified Mentor contains customer orders, shipment information, product details, sales records, factory locations, and customer geographical information. The dataset served as the primary source for all analytical operations performed in this project.

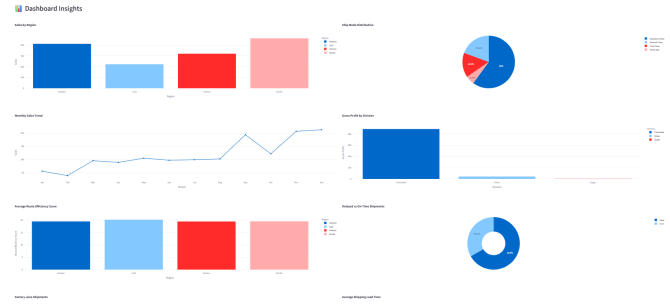


Fig. 2. Interactive Streamlit Dashboard for Logistics Performance Analysis

B. Data Cleaning and Validation

Data preprocessing was performed before analysis to ensure consistency and improve data quality. The following operations were carried out:

- Conversion of Order Date and Ship Date into datetime format.
- Identification and removal of invalid or inconsistent records.
- Verification of missing values.
- Standardisation of region and division names.
- Validation of numerical columns including Sales, Cost and Gross Profit.
- Removal of records with invalid shipping lead times.

These preprocessing steps ensured reliable analytical results throughout the dashboard.

C. Feature Engineering

Several additional variables were created from the original dataset to improve logistics analysis.

The engineered features include:

- **Shipping Lead Time**
Calculated as:

$$\text{Shipping Lead Time} = \text{Ship Date} - \text{Order Date}$$

- **Route Efficiency Score**
A derived metric used to compare shipping routes based on average delivery performance.
- **Factory-to-Region Route**
Generated for analysing regional shipment efficiency.
- **Factory-to-State Route**
Used for detailed geographical route analysis.
- **Delay Frequency**
Calculated to identify shipments exceeding the expected delivery duration.

Feature engineering significantly enhanced the analytical capability of the dashboard.

D. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand shipment behaviour, identify trends, detect anomalies, and evaluate logistics performance.

The following analyses were conducted:

- Regional Sales Analysis
- Gross Profit Analysis
- Shipping Lead Time Distribution
- Factory Performance
- Route Efficiency Analysis
- Ship Mode Comparison
- Shipment Volume Analysis

The results obtained from EDA formed the foundation for dashboard visualisation and business insight generation.

E. KPI Development

Key Performance Indicators (KPIs) were developed to summarise business performance.

The dashboard calculates:

- Total Shipments
- Total Sales
- Gross Profit
- Average Shipping Lead Time
- Best Performing Region
- Highest Profit Division
- Fastest Shipping Mode

These KPIs provide management with an instant overview of organisational logistics performance.

F. Dashboard Development

The final phase involved designing an interactive dashboard using Streamlit.

The dashboard integrates:

- Interactive Filters
- KPI Cards
- Plotly Charts
- Product Search
- Download Filtered Dataset
- Business Insights

The dashboard enables decision-makers to analyse logistics performance dynamically through interactive visualisations and real-time filtering capabilities.

VII. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) was conducted to examine shipment behaviour, identify business trends, evaluate logistics performance, and generate actionable insights. Various statistical summaries and interactive visualisations were created using Plotly and Streamlit.

The primary objective of EDA was to transform raw shipment data into meaningful business intelligence that supports operational decision-making.

A. Regional Sales Analysis

Regional sales analysis was performed to identify the contribution of each customer region towards the overall revenue generated by Nassau Candy Distributor.

A bar chart was developed to compare total sales across different regions. This analysis helps management identify high-performing markets and regions requiring additional business focus.

Key Insight:

- Regions with higher sales contribute significantly towards organisational revenue.
- Regional comparison assists in strategic planning and resource allocation.

B. Gross Profit Analysis

Gross profit was analysed to evaluate the profitability generated across different business divisions and customer regions.

This analysis enables decision-makers to distinguish between high-sales regions and high-profit regions, which may not always be the same.

Key Insight:

- Profitability provides a better measure of business performance than sales alone.
- Profit analysis supports financial planning and pricing strategies.

C. Shipping Lead Time Analysis

Shipping Lead Time represents the number of days required to deliver customer orders.

The lead time was calculated using the difference between Ship Date and Order Date.

This metric serves as one of the most important logistics KPIs because it directly impacts customer satisfaction and operational efficiency.

Key Insight:

- Lower lead times indicate better logistics performance.
- Higher lead times may indicate transportation bottlenecks or inefficient shipping routes.

D. Ship Mode Performance Analysis

Different shipping methods were compared to evaluate delivery efficiency.

The analysis compares average shipping lead time across available ship modes.

Key Insight:

- Faster shipping modes generally provide shorter delivery durations.
- The comparison assists in selecting the most efficient transportation strategy.

E. Factory Performance Analysis

Shipment volume and business performance were analysed across all manufacturing facilities.

Factory-level analysis helps identify facilities handling the highest shipment volume and contributing significantly towards sales and profit.

Key Insight:

- Factory performance analysis assists in workload balancing.
- Efficient factories can serve as benchmarks for operational improvement.

F. Route Efficiency Analysis

Route efficiency was analysed by comparing shipment performance across different customer regions and states.

The objective was to identify both efficient and inefficient logistics routes.

Key Insight:

- High-performing routes indicate efficient logistics operations.
- Poor-performing routes require operational investigation and optimisation.

G. Dashboard Insights

The interactive dashboard allows users to perform dynamic analysis through filters and visualisations.

The implemented dashboard supports:

- Region-wise analysis
- Division-wise analysis
- Ship Mode comparison
- Product search
- Interactive KPI monitoring
- Downloading filtered datasets

The dashboard transforms static business reports into an interactive analytics platform that enables faster and more informed decision-making.

VIII. DASHBOARD DEVELOPMENT

The final phase of the project involved developing an interactive web-based analytics dashboard using Streamlit. The dashboard was designed to transform processed logistics data into an intuitive decision-support system capable of providing real-time business insights through interactive visualisations and user-controlled filters.

Python was selected as the primary programming language due to its extensive ecosystem for data analysis and visualisation. Streamlit was chosen as the application framework because it enables rapid development of interactive dashboards without requiring front-end web development. Plotly Express was used to create dynamic and responsive charts that update instantly based on user selections.

The dashboard follows a modular architecture where each analytical component is implemented independently and integrated into a unified interface. Separate Python modules were developed for data processing, route analysis, KPI calculation, feature engineering, and ship mode analysis. This modular design improves code readability, maintainability, and scalability.

A. Dashboard Architecture

The overall workflow of the dashboard is illustrated below.

- 1) Load the cleaned shipment dataset.
- 2) Apply user-selected filters.
- 3) Calculate business KPIs.
- 4) Generate interactive Plotly visualisations.
- 5) Display insights through the Streamlit interface.
- 6) Allow users to download filtered datasets.

This architecture ensures efficient processing while maintaining an interactive user experience.

B. Interactive Sidebar

The dashboard includes an interactive sidebar that allows users to customise the displayed analytics according to their business requirements.

The implemented filters include:

- Region Filter
- Division Filter
- Ship Mode Filter
- Product Name Search
- Download Filtered Dataset

Whenever the user changes any filter, all charts and KPIs automatically refresh to display the updated analytical results.

C. Key Performance Indicators

A KPI section was developed to provide a quick overview of logistics performance.

The dashboard displays the following metrics:

- Total Shipments
- Total Sales
- Gross Profit
- Average Shipping Lead Time

These indicators provide management with an immediate understanding of organisational performance without requiring detailed report analysis.

D. Interactive Visualisations

Several interactive Plotly charts were incorporated into the dashboard to simplify business analysis.

The dashboard includes:

- Sales by Region
- Gross Profit Analysis
- Ship Mode Comparison
- Factory Performance
- Shipment Distribution
- Route Efficiency Analysis

Each chart dynamically responds to the selected dashboard filters, allowing users to perform customised exploratory analysis.

E. Business Insights Panel

To improve decision-making, the dashboard automatically highlights important business insights based on the filtered data.

Examples include:

- Best Performing Sales Region
- Highest Profit Division
- Fastest Shipping Mode

These automatically generated insights help users quickly identify important business trends.

F. Technology Stack

The technologies used throughout the project are summarised in Table III.

The developed dashboard successfully converts historical shipment records into an interactive business intelligence platform capable of supporting logistics optimisation and operational decision-making.

IX. RESULTS AND DISCUSSION

The developed analytics dashboard successfully transformed raw shipment data into meaningful logistics insights. The analysis performed through data preprocessing, feature engineering, and interactive visualisation enabled a detailed evaluation of shipping performance, regional trends, and operational efficiency.

The results obtained from the dashboard demonstrate the effectiveness of data analytics in identifying patterns within large-scale shipment operations and supporting data-driven decision-making.

A. Shipping Performance Analysis

Shipping Lead Time was identified as one of the most important indicators for evaluating logistics efficiency. By calculating the duration between order placement and shipment delivery, the dashboard provides visibility into delivery performance across different regions and shipping methods.

The analysis enables users to identify routes with shorter delivery durations and detect areas where delays frequently occur. This information can help logistics teams investigate the reasons behind delayed shipments and improve transportation planning.

B. Regional Performance Analysis

The regional analysis provided insights into variations in sales contribution, shipment volume, and logistics performance across different customer regions.

The dashboard enables comparison between regions based on:

- Total number of shipments
- Sales generated
- Gross profit contribution
- Average shipping lead time

This comparison helps identify high-performing regions as well as regions requiring operational improvements.

C. Ship Mode Evaluation

The comparison of different shipping methods provided insights into the relationship between transportation mode and delivery efficiency.

The dashboard allows users to evaluate:

- Average lead time for each shipping mode
- Shipment distribution across shipping methods
- Efficiency differences between standard and faster delivery options

This analysis supports better decision-making regarding the selection of shipping methods while balancing delivery speed and operational costs.

D. Factory and Route Performance

Factory-level and route-level analysis provided visibility into the relationship between manufacturing locations and customer delivery destinations.

The developed solution helps identify:

- Efficient factory-to-region routes
- High-volume shipment routes
- Potential logistics bottlenecks

Understanding route performance allows organisations to improve resource allocation and optimise transportation strategies.

E. Business Intelligence Benefits

The interactive nature of the Streamlit dashboard provides several advantages compared with traditional static reports.

The major benefits include:

- Faster identification of logistics issues.
- Improved visibility into shipment performance.
- Reduced dependency on manual reporting.
- Easy comparison of operational metrics.
- Support for data-driven business decisions.

Overall, the project demonstrates that applying data analytics techniques to shipment data can significantly improve understanding of logistics operations. The developed dashboard acts as a decision-support tool that enables users to monitor performance, identify improvement opportunities, and plan effective optimisation strategies.

X. BUSINESS RECOMMENDATIONS

Based on the insights obtained from the data analysis and visualization of shipping operations, the following business recommendations are proposed to improve logistics efficiency and operational performance:

- **Optimize Shipping Routes:** The analysis of route efficiency scores and shipping lead times indicates that certain routes require optimization. The company should prioritize high-efficiency routes and redesign low-performing routes to reduce transportation delays and improve delivery performance.
- **Improve Delivery Time Management:** Regions and shipping modes with higher delivery lead times should be monitored closely. Implementing better scheduling strategies and selecting appropriate shipping methods can help reduce delays and increase customer satisfaction.
- **Adopt Data-Driven Inventory Planning:** Sales trends and product performance analysis can be used to forecast demand more accurately. This enables better inventory allocation, reduces stock shortages, and minimizes excess inventory costs.
- **Enhance Shipping Mode Selection:** The company should evaluate the performance of different shipping modes based on cost, delivery speed, and customer requirements. Selecting the most suitable shipping mode for each order can improve overall logistics efficiency.
- **Monitor Regional Performance:** Regional analysis highlights variations in sales, profitability, and delivery efficiency. Regular monitoring of regional KPIs can help identify improvement opportunities and support strategic decision-making.

- **Implement Continuous Performance Monitoring:** The developed Streamlit dashboard can be used as a decision-support tool for real-time monitoring of logistics metrics. Continuous tracking of key indicators such as shipping lead time, profit, and route efficiency can help managers make informed decisions.

The implementation of these recommendations can help organizations reduce operational inefficiencies, improve supply chain performance, and achieve higher customer satisfaction through effective data-driven decision-making.

XI. CONCLUSION

This research presents a data-driven approach for analyzing and improving logistics operations using data analytics techniques. The study focuses on identifying patterns in shipping performance, delivery efficiency, sales trends, and route effectiveness using the Nassau Candy Distributor dataset.

The developed analytics framework utilizes data preprocessing, exploratory data analysis, and interactive visualization techniques to transform raw operational data into meaningful insights. The Streamlit-based dashboard provides an effective platform for monitoring key performance indicators such as shipping lead time, route efficiency, regional performance, sales distribution, and profitability.

The analysis demonstrates that data analytics can significantly support decision-making in supply chain management by identifying operational bottlenecks and improvement opportunities. The insights generated from the dashboard can help organizations optimize shipping routes, select appropriate transportation modes, reduce delivery delays, and improve overall customer satisfaction.

Overall, this work highlights the importance of integrating data analytics tools into logistics management systems. By adopting data-driven strategies, businesses can enhance operational efficiency, reduce costs, and achieve better supply chain performance.

XII. FUTURE SCOPE

The proposed data analytics framework can be further enhanced by incorporating advanced technologies and real-time data processing capabilities. Future improvements can focus on developing more intelligent and automated logistics management solutions.

- **Integration of Machine Learning Models:** Predictive machine learning algorithms can be incorporated to forecast delivery delays, demand patterns, shipping costs, and future sales trends. This can enable proactive decision-making and improve operational planning.
- **Real-Time Data Integration:** Future versions of the system can integrate real-time tracking data from GPS and IoT-enabled devices to monitor shipments continuously and provide live updates on delivery status.
- **Advanced Predictive Analytics:** Implementing advanced analytics techniques can help predict potential supply chain risks, optimize inventory levels, and improve resource allocation.

- **Automated Decision Support System:** The dashboard can be extended into an automated recommendation system that provides optimized shipping routes, cost-saving strategies, and operational suggestions based on historical and real-time data.
- **Scalability with Larger Datasets:** The proposed approach can be tested on larger and more diverse logistics datasets from multiple industries to improve generalization and scalability.
- **Cloud-Based Deployment:** Deploying the analytics platform on cloud infrastructure can improve accessibility, collaboration, and scalability for organizations requiring enterprise-level logistics monitoring.

By incorporating these advancements, the proposed system can evolve from a descriptive analytics dashboard into a comprehensive intelligent logistics optimization platform capable of supporting automated and predictive decision-making.

REFERENCES

- [1] J. Han, M. Kamber, and J. Pei, *Data Mining: Concepts and Techniques*, 3rd ed. San Francisco, CA, USA: Morgan Kaufmann, 2012.
- [2] R. Kimball and M. Ross, *The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modeling*, 3rd ed. Indianapolis, IN, USA: Wiley, 2013.
- [3] T. M. Choi, "Supply chain management in the era of big data analytics," *International Journal of Production Economics*, vol. 193, pp. 1–3, 2017.
- [4] P. R. Murphy, D. J. Daley, and D. L. Dalenberg, "Logistics performance and firm success: An empirical study," *Journal of Business Logistics*, vol. 20, no. 2, pp. 29–48, 1999.
- [5] W. McKinney, *Python for Data Analysis: Data Wrangling with Pandas, NumPy, and Jupyter*, 3rd ed. Sebastopol, CA, USA: O'Reilly Media, 2022.
- [6] P. Virtanen *et al.*, "Scipy 1.0: Fundamental algorithms for scientific computing in Python," *Nature Methods*, vol. 17, pp. 261–272, 2020.
- [7] M. Zaharia *et al.*, "Apache Spark: A unified engine for big data processing," in *Proceedings of the 9th USENIX Conference on Hot Topics in Cloud Computing*, 2016.
- [8] Streamlit Inc., "Streamlit: The fastest way to build data apps," 2024. [Online]. Available: <https://streamlit.io/>
- [9] The Pandas Development Team, "Pandas documentation," 2024. [Online]. Available: <https://pandas.pydata.org/>
- [10] Plotly Technologies Inc., "Plotly Python Graphing Library," 2024. [Online]. Available: <https://plotly.com/python/>

TABLE I
MANUFACTURING FACTORY COORDINATES

Factory	Latitude	Longitude
Lot's O' Nuts	32.881893	-111.768036
Wicked Choccy's	32.076176	-81.088371
Sugar Shack	48.119140	-96.181150
Secret Factory	41.446333	-90.565487
The Other Factory	35.117500	-89.971107

TABLE II
PRODUCT MANUFACTURING LOCATIONS

Product Division	Manufacturing Factory
Chocolate Products	Lot's O' Nuts / Wicked Choccy's
Sugar Products	Sugar Shack / Secret Factory
Other Products	Secret Factory / The Other Factory

TABLE III
TECHNOLOGY STACK

Technology	Purpose
Python	Data Processing
Pandas	Data Manipulation
NumPy	Numerical Operations
Plotly Express	Interactive Charts
Streamlit	Dashboard Development
Git	Version Control
GitHub	Project Repository
VS Code	Development Environment